Application Network Diagram

Ref: 1.3
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: Network operations, security assessors, platform engineering

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1. Purpose

Document the **on-premises network placement** of iConnect Plus: VLAN/LAN segments, VPN entry, reverse proxy, Docker application containers, and data-tier segregation. This document **does not** describe an AWS VPC or public multi-tenant SaaS edge as the primary model.

2. Network overview

Segment	Typical VLAN / subnet	Systems	Access
Corp user LAN	Office user subnet	Workstations	Domain / corp auth
Server LAN	Internal server subnet (e.g. `172.16.200.0/24`)	Application Docker host	Restricted ACL
Management	IT admin subnet (optional)	SSH, backup, monitoring	Jump host / privileged
VPN pool	Remote client addresses	VPN clients	MFA on VPN (**Template / Pending sign-off**)

Example application host referenced in operations: **`172.16.200.30`**.

3. Entry paths

3.1 Office users

```
Workstation (Corp LAN) ──HTTPS──► 172.16.200.30:8443 (nginx) ──► web / api containers
```

3.2 Remote users

```
Remote device ──VPN──► Corp LAN routing ──HTTPS──► 172.16.200.30:8443
```

Control	Requirement
Public internet exposure	Application **not** required to be on public internet
VPN	**Mandatory** for out-of-office access
Split tunneling	**Template / Pending sign-off** (IT policy)

3.3 Administrators

Function	Path	Notes
OS / Docker admin	SSH to host from IT network	Key-based; **Template / Pending sign-off**
DB admin	`psql` via LAN or port-forward	Not from untrusted networks
MinIO console	Internal `:9001`	LAN/VPN only
RabbitMQ management	Internal `:15672`	Restricted

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4. On-host container topology

Docker bridge network: **`erp-net`**

Container	Service	Published ports (example prod coexist)	Internal DNS
`erp-nginx`	TLS reverse proxy	`8443→443`	`nginx`
`erp-web`	Next.js admin	`3000` (optional host bind)	`web`
`erp-employee-app`	ESS	`3001`	`employee-app`
`erp-api`	FastAPI	`8000`	`api`
`erp-celery-worker`	Background jobs	—	—
`erp-postgres`	PostgreSQL	`5433→5432` (host coexist)	`postgres`
`erp-redis`	Redis	`6379`	`redis`
`erp-rabbitmq`	RabbitMQ	`5672`, `15672`	`rabbitmq`
`erp-minio`	Object storage	`9000`, `9001`	`minio`
`erp-clamav`	Antivirus	`3310`	`clamav`
`erp-opensearch`	Search (optional)	`9200`	`opensearch`

Compose files: `docker-compose.infra.yml` (data tier) + `docker-compose.app.yml` (application tier).

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5. Segregation controls

Control	Implementation	Owner
LAN firewall	Allow 8443 (and admin ports) only from corp + VPN ranges	Cache Digitech IT
Container isolation	Separate containers; no direct DB from browsers	Platform
East-west	API only talks to DB on `erp-net`	Platform
TLS termination	Nginx with org certs (`docker/nginx/certs/`)	IT + Platform
No public SaaS dependency	MinIO + Postgres on same premises host	Platform

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6. DNS and URLs (internal)

Role	Example value	Notes
Admin UI base	`https://172.16.200.30:8443`	Internal IP or internal DNS name
API (browser)	`https://172.16.200.30:8443/api/v1`	Via nginx

API (container)	`http://api:8000/api/v1`	`API_INTERNAL_URL`
MinIO S3 API	`http://minio:9000`	`S3_ENDPOINT_URL`

Internal DNS records (if used): **Template / Pending sign-off**

7. Logical network diagram

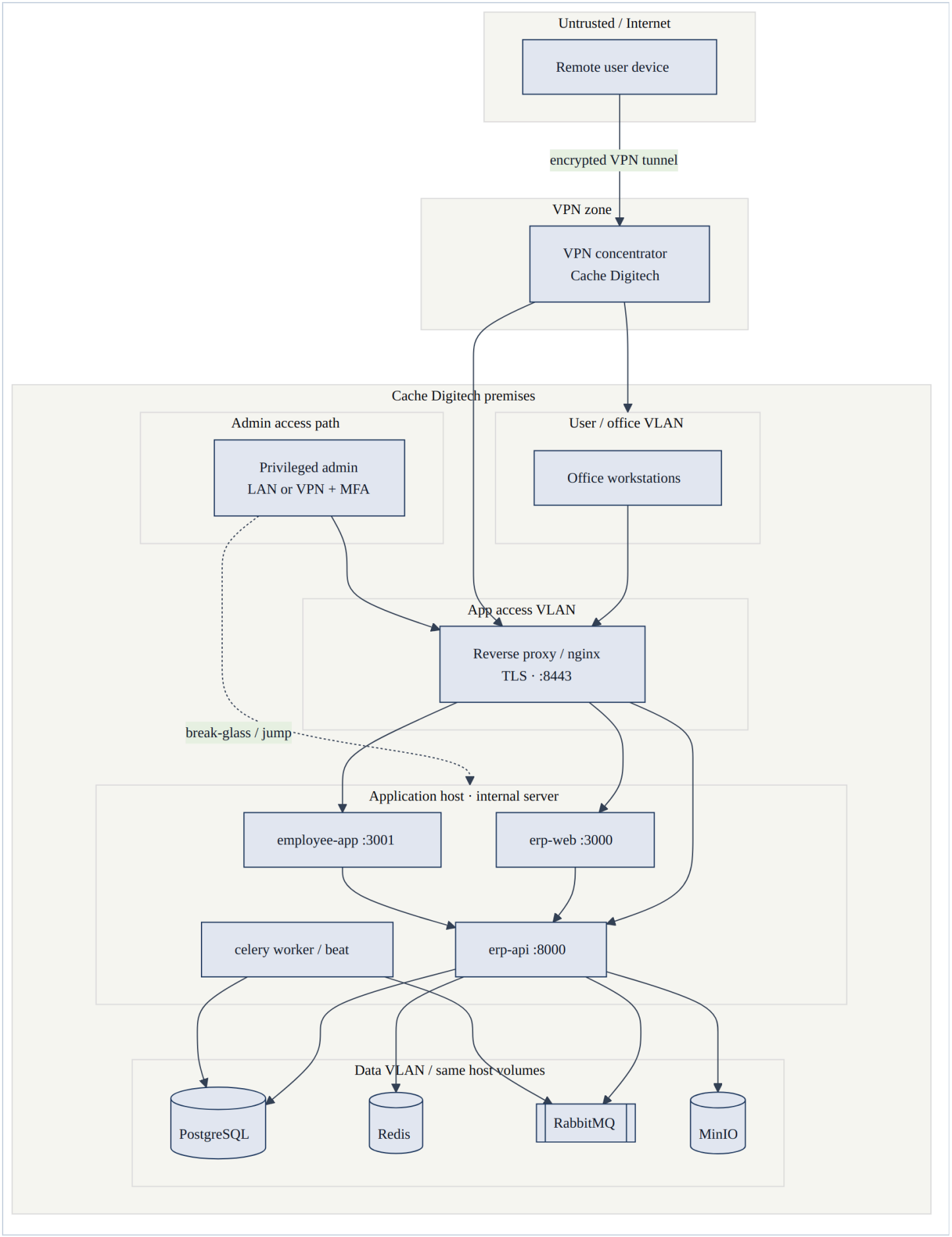


Figure 1 — Cache Digitech premises network: remote users must traverse the VPN concentrator before reaching the corporate LAN / app VLAN; office users reach the reverse proxy on the internal server; all frontend and backend containers and data stores run on-premises.

8. Ports summary (firewall checklist)

Port	Service	Source allowed
8443	HTTPS app	Corp LAN + VPN pools
8000	API direct (dev)	Restrict in prod — prefer nginx only
5433/5432	PostgreSQL	Localhost / admin jump only
9000/9001	MinIO	Internal admin
22	SSH	IT management subnet

Production hardening evidence: Template / Pending sign-off

9. Related documents

Ref	Document
1.1	Solution Architecture
1.2	Data Flow Diagram
1.5	Asset & Component Inventory
1.7	Infrastructure Shared-Responsibility Matrix

10. Document control

Version	Date	Change
2.0	2026-11-03	Premises LAN/VPN model

Approval: Template / Pending sign-off